AI-Based Cytology Image Classification Under Limited Data Conditions

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***Abstract*—This project investigates the feasibility of training deep learning models for cervical cytology classification under limited data conditions. A dataset of approximately 600 images from the BMT ThinPrep Pap smear collection was used to classify samples into three categories: NILM, LSIL, and HSIL. Multiple deep learning approaches were explored, including convolutional neural networks, pretrained architectures, and a CLIP-based model. Models were evaluated using accuracy, F1-score, and AUC. Results show that transfer learning significantly improves performance, with ResNet achieving the highest F1-score.**

I. Introduction

Medical imaging datasets, particularly in cytology, are often constrained in size, which poses challenges for developing high-performance models. This study explores multiple deep learning approaches for classifying cervical cytology images into NILM, LSIL, and HSIL categories. The primary objective is to approach an F1-score of 90% despite limited data.

II. Dataset and Methods

A. Dataset

The dataset used is the BMT ThinPrep Pap smear cytology dataset, containing labeled cervical cell images categorized into three classes: NILM, LSIL, and HSIL. A key challenge is that these classes share many visual similarities, differing mainly in cellular morphology.

B. Data Pipeline

Images were resized to 224×224 pixels and normalized using pretrained model statistics. The dataset was split into training and validation sets. Data augmentation was applied to artificially increase dataset size.

C. Models

VGG Model: A pretrained VGG network was used as a feature extractor with a custom fully connected classification head.

ResNet Model: A pretrained ResNet architecture with residual connections was fine-tuned with a custom classifier.

Control CNN: A baseline convolutional neural network trained from scratch to compare performance.

CLIP Model: A pretrained vision encoder was used to extract embeddings, followed by a custom classification head. Partial freezing and fine-tuning were applied.

D. Training Setup

The Adam optimizer was used for training. All models were trained with:

- Batch size: 16
- Epochs: 30
- Loss: Sparse categorical cross-entropy

A layer-wise learning rate strategy was used for the CLIP model.

E. Evaluation Metrics

Performance was evaluated using Accuracy, F1-score, and AUC. F1-score was prioritized due to class imbalance.

III. Results

- **Control CNN:** Accuracy: 30%, F1: 13.84%, AUC: 50%
- **ResNet:** Accuracy: 90.83%, F1: 90.82%, AUC: 96.90%
- **VGG:** Accuracy: 82.5%, F1: 82.54%, AUC: 94.89%
- **CLIP:** Accuracy: 83.75%, F1: 83.56%, AUC: 94.91%

IV. Discussion

All pretrained models significantly outperformed the control CNN, demonstrating the effectiveness of transfer learning in limited data scenarios. ResNet achieved the best performance, likely due to its residual connections enabling deeper feature extraction.

The CLIP-based model required additional customization and did not perform as consistently, suggesting that foundation models may not directly generalize to specialized medical imaging tasks without adaptation.

A key limitation across models is the difficulty in capturing fine morphological differences between classes.

V. Conclusion

This study demonstrates that transfer learning is essential for cytology classification under limited data conditions. ResNet50 achieved the highest performance, indicating the advantage of deeper architectures.

Future work should focus on increasing dataset size and incorporating attention mechanisms or region-focused training to better capture morphological features.